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Subject: Mathematics | **CBSE CLASS 10 – CHAPTER TEST (SET – 9)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM: 20**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. The distance of the point $P(-6, 8)$ from the origin is: [1]

(A) 8 **(B)** $2\sqrt{7}$ **(C)** 10 **(D)** 6

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 10

Given: Point $P(-6, 8)$ and origin $O(0, 0)$.

Formula Used: $OP = \sqrt{x^2 + y^2}$

$$OP = \sqrt{(-6)^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = \boxed{\textbf{10 units}}$$

Q2. The distance of which of the following points from origin is less than 5 units? [1]

(A) $(3, 4)$ **(B)** $(2, 6)$ **(C)** $(-3, -4)$ **(D)** $(1, 4)$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D) $(1, 4)$

Concept: Distance of $P(x, y)$ from origin $O(0, 0)$ is strictly less than 5 $\iff$ $\sqrt{x^2 + y^2} < 5$.

Formula Used: $d = \sqrt{x^2 + y^2}$

Checking all four options:

- $(3, 4) \implies \sqrt{3^2 + 4^2} = \sqrt{25} = 5$ (equal to 5, not less).
- $(2, 6) \implies \sqrt{2^2 + 6^2} = \sqrt{40} > 5$.
- $(-3, -4) \implies \sqrt{(-3)^2 + (-4)^2} = \sqrt{25} = 5$ (equal to 5, not less).
- $(1, 4) \implies \sqrt{1^2 + 4^2} = \sqrt{17} \approx 4.12 < 5 \implies \boxed{\textbf{(1, 4)}}$.

Q3. The distance of point $P(1,-1)$ from the x-axis is: [1]

(A) 1 **(B)** -1 **(C)** 0 **(D)** $\sqrt{2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 1

Given: Point $P(1,-1)$. Foot of perpendicular on the x-axis is $M(1,0)$.

Formula Used: Distance of (x,y) from x-axis $= |y|$

$$PM = \sqrt{(1-1)^2 + (0-(-1))^2} = \sqrt{1} = |-1| = \boxed{\mathbf{1}\text{ unit}}$$

Q4. The point on the x-axis which is equidistant from the points $A(-1,0)$ and $B(5,0)$ is: [1]

(A) $(2,0)$ **(B)** $(0,2)$ **(C)** $(3,0)$ **(D)** $(-2,0)$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $(2,0)$

Given: Both $A(-1,0)$ and $B(5,0)$ lie on the x-axis; their equidistant point on the x-axis is their midpoint $P(x,0)$.

Formula Used: $P = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$$P = \left(\frac{-1+5}{2}, \frac{0+0}{2}\right) = \left(\frac{4}{2}, 0\right) = \boxed{\mathbf{(2,0)}}$$

Q5. Find the distance between the points $A(0,6)$ and $B(0,-2)$. [1]

(A) 6 units **(B)** 8 units **(C)** 4 units **(D)** 10 units

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 8 units

Given: Points $A(0,6)$ and $B(0,-2)$.

Formula Used: $AB = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

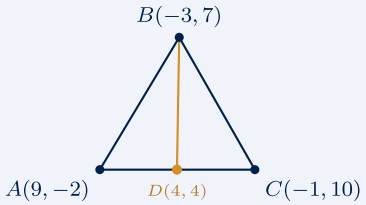
$$AB = \sqrt{(0-0)^2 + (-2-6)^2} = \sqrt{0+(-8)^2} = \sqrt{64} = \boxed{\mathbf{8}\text{ units}}$$

SECTION B: Short Answer Type Questions

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Q6. Find the length of the median through the vertex B of $\triangle ABC$ with vertices $A(9,-2)$, $B(-3,7)$ and $C(-1,10)$. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Finding Midpoint D of Side AC [1 Mark]
Let the median through $B(-3,7)$ meet opposite side AC at its midpoint D, where $A(9,-2)$ and $C(-1,10)$.

We know the midpoint and distance formulas:

Formula Used: $D = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$BD = \sqrt{(x_D - x_B)^2 + (y_D - y_B)^2}$

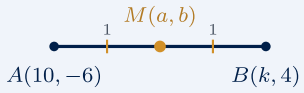
$$D = \left(\frac{9+(-1)}{2}, \frac{-2+10}{2}\right) = \left(\frac{8}{2}, \frac{8}{2}\right) = (4,4)$$

Step 2: Length of Median BD [1 Mark]

$$BD = \sqrt{(4-(-3))^2 + (4-7)^2} = \sqrt{7^2 + (-3)^2} = \sqrt{49+9} = \boxed{\sqrt{58} \text{ units}}$$

Q7. If (a, b) is the mid-point of the line segment joining the points $A(10, -6)$ and $B(k, 4)$ and $a - 2b = 18$, then find the value of k. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Midpoint Coordinates in Terms of k [1 Mark]
Given: Endpoints $A(10,-6)$ and $B(k,4)$, midpoint (a,b), and linear condition $a - 2b = 18$.

We know the midpoint formula:

Formula Used: $(a, b) = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$$a = \frac{10+k}{2}, \qquad b = \frac{-6+4}{2} = \frac{-2}{2} = -1$$

Step 2: Substituting into $a - 2b = 18$ to Find k [1 Mark]

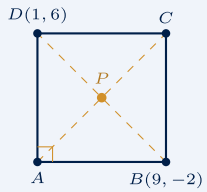
$$\frac{10+k}{2} - 2(-1) = 18 \implies \frac{10+k}{2} + 2 = 18 \implies \frac{10+k}{2} = 16$$

$$10 + k = 32 \implies k = 32 - 10 = \boxed{\mathbf{22}}$$

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Q8. Diagonals AC and BD of square $ABCD$ intersect at P. Coordinates of points B and D are $(9, -2)$ and $(1, 6)$ respectively. (i) Find the co-ordinates of point P. (ii) Find the length of the side of the square. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



(i) Finding Intersection P (Midpoint of BD): [1 Mark]
Since the diagonals of a square bisect each other, P is the midpoint of diagonal BD joining $B(9, -2)$ and $D(1, 6)$.

We know the midpoint and diagonal properties of a square:

Formula Used: $P = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$, Diagonal $= s\sqrt{2}$

$$P = \left(\frac{9+1}{2}, \frac{-2+6}{2}\right) = \left(\frac{10}{2}, \frac{4}{2}\right) = (\mathbf{5}, \mathbf{2})$$

(ii) Finding Length of Side of the Square (s): [1 Mark]
First find diagonal $BD = \sqrt{(1-9)^2 + (6-(-2))^2} = \sqrt{(-8)^2 + 8^2} = \sqrt{64+64} = \sqrt{128} = 8\sqrt{2}$ units.
Since diagonal of a square = side $\times \sqrt{2}$ (by Pythagoras' theorem $s^2 + s^2 = BD^2$):

$$s\sqrt{2} = 8\sqrt{2} \implies \mathbf{s = 8} \text{ units}$$

$P = (\mathbf{5}, \mathbf{2})$ and Side of the square $= \mathbf{8}$ units

SECTION C: Long Answer & Competency Based Questions (9 Marks)

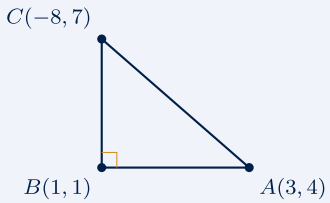
Q9. Vertices of a right triangle ABC with $\angle B = 90°$ are $A(3, 4)$, $B(1, 1)$ and $C(-8, 7)$. Find the value of $\tan A$. **[5]**

— OR —

Using distance formula, prove that the points $A(2, 3)$, $B(-7, 0)$ and $C(-1, 2)$ are collinear. **[5]**

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▶ **SOLUTION (OPTION I):** [Total: 5 Marks]



Step 1: Finding Side Lengths AB and BC [3 Marks]
Given: Right $\triangle ABC$ with $\angle B = 90^\circ$ and vertices $A(3,4)$, $B(1,1)$, and $C(-8,7)$.

We know the distance formula and trigonometric ratio for a right triangle:

Formula Used: $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$, $\tan A = \dfrac{\text{Opposite side } (BC)}{\text{Adjacent side } (AB)}$

Computing lengths:

$$AB = \sqrt{(1-3)^2+(1-4)^2} = \sqrt{(-2)^2+(-3)^2} = \sqrt{4+9} = \sqrt{\mathbf{13}} \text{ units}$$

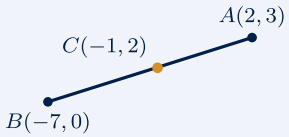
$$BC = \sqrt{(-8-1)^2+(7-1)^2} = \sqrt{(-9)^2+6^2} = \sqrt{81+36} = \sqrt{117} = \mathbf{3\sqrt{13}} \text{ units}$$

(Checking $\angle B = 90^\circ$: $AC = \sqrt{(-8-3)^2+(7-4)^2} = \sqrt{130}$, and $AB^2 + BC^2 = 13 + 117 = 130 = AC^2$.)

Step 2: Calculating $\tan A$ [2 Marks]

$$\tan A = \frac{BC}{AB} = \frac{3\sqrt{13}}{\sqrt{13}} = \boxed{\mathbf{3}}$$

▶ **SOLUTION (OPTION II):** [OR: 5 Marks]



Step 1: Calculating All Three Pairwise Distances [3.5 Marks]
Given: Points $A(2,3)$, $B(-7,0)$, and $C(-1,2)$. Three points are collinear if the sum of the two shorter distances equals the longest distance.

We know the distance formula:

Formula Used: $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$$AB = \sqrt{(-7-2)^2+(0-3)^2} = \sqrt{(-9)^2+(-3)^2} = \sqrt{81+9} = \sqrt{90} = \mathbf{3\sqrt{10}} \text{ units}$$

$$BC = \sqrt{(-1-(-7))^2+(2-0)^2} = \sqrt{6^2+2^2} = \sqrt{36+4} = \sqrt{40} = \mathbf{2\sqrt{10}} \text{ units}$$

$$AC = \sqrt{(-1-2)^2+(2-3)^2} = \sqrt{(-3)^2+(-1)^2} = \sqrt{9+1} = \sqrt{\mathbf{10}} \text{ units}$$

Step 2: Verifying Distance Sum Condition [1.5 Marks]
Checking the sum of the two smaller lengths against the longest distance:

$$AC + BC = \sqrt{10} + 2\sqrt{10} = 3\sqrt{10} = AB$$

$$\boxed{\text{Since } AC + BC = AB, \text{ points } \mathbf{A}, \mathbf{B}, \mathbf{C} \text{ are collinear (Hence Proved)}}$$

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Q10. Competency Case Study: Observe the map of Jaipur city placed on a Cartesian plane. Taking Rambagh Palace as origin $O(0,0)$, the locations of some places are given below: Point $A(-4,2)$ Rajasthan High Court, Point $B(4,-4)$ Birla Mandir, Point $C(4,3)$ Heera Bagh, Point $D(-5,-2)$ Amar Jawan Jyoti. **[4]**

(i) Advocate Rehana stays at Heera Bagh. How much distance does she have to cover daily to go to the court and come back home? **[1 Mark]**

(ii) There is a crossing on the x-axis which divides AD in a certain ratio. Find the ratio. **[1 Mark]**

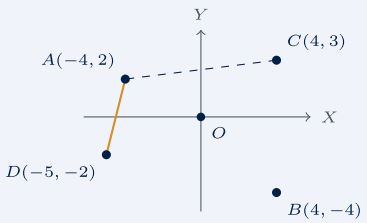
(iii) **(a)** Is Birla Mandir equidistant from Heera Bagh and Amar Jawan Jyoti? Justify your answer. **[2 Marks]**

— OR —

(iv) **(b)** Using section formula, show that points A, O and B are not collinear. **[2 Marks]**

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▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Given: Origin Rambagh Palace $O(0,0)$, High Court $A(-4,2)$, Birla Mandir $B(4,-4)$, Heera Bagh $C(4,3)$, and Amar Jawan Jyoti $D(-5,-2)$. We apply the distance and internal section formulas.

We know the distance and internal section formulas:

Formula Used:
$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$
$$P(x,y) = \left(\frac{m_1x_2+m_2x_1}{m_1+m_2}, \frac{m_1y_2+m_2y_1}{m_1+m_2}\right)$$

(i) Daily Round-Trip Distance ($C \leftrightarrow A$): [1 Mark]

One-way distance from Heera Bagh $C(4,3)$ to High Court $A(-4,2)$:

$$CA = \sqrt{(-4-4)^2+(2-3)^2} = \sqrt{(-8)^2+(-1)^2} = \sqrt{64+1} = \sqrt{65} \text{ units}$$

Daily distance (going + returning) $= 2 \times CA = \mathbf{2\sqrt{65}}$ units.

(ii) Ratio in Which x-Axis Crossing Divides AD: [1 Mark]

Let $(x,0)$ on x-axis divide $A(-4,2)$ and $D(-5,-2)$ in ratio $k:1$. Setting y-coordinate to 0:

$$\frac{k(-2)+1(2)}{k+1} = 0 \implies -2k+2=0 \implies k=1 \implies \text{Ratio} = \mathbf{1:1}$$

(iii) (a) Checking If Birla Mandir $B(4,-4)$ is Equidistant from $C(4,3)$ and $D(-5,-2)$: [2 Marks]

$$BC = \sqrt{(4-4)^2+(3-(-4))^2} = \sqrt{0+7^2} = \mathbf{7} \text{ units}$$

$$BD = \sqrt{(-5-4)^2+(-2-(-4))^2} = \sqrt{(-9)^2+2^2} = \sqrt{81+4} = \sqrt{\mathbf{85}} \text{ units}$$

Since $BC \neq BD$ ($7 \neq \sqrt{85}$), **No**, Birla Mandir is not equidistant from them.

— OR —

(iv) (b) Showing $A(-4,2)$, $O(0,0)$, $B(4,-4)$ Not Collinear via Section Formula: [OR: 2 Marks]

Suppose $O(0,0)$ divides segment AB in ratio $k:1$. Equating the x-coordinate to 0:

$$\frac{k(4)+1(-4)}{k+1} = 0 \implies 4k-4=0 \implies k=1$$

For $k=1$, the y-coordinate of the dividing point on AB is:

$$y = \frac{1(-4)+1(2)}{1+1} = \frac{-2}{2} = -1 \neq 0$$

Since $(0,0) \neq (0,-1)$, O does not lie on AB; hence A, O, and B are **not collinear**.

$\boxed{(i)\, 2\sqrt{65} \text{ units};\ \ (ii)\, 1:1;\ \ (a)\, \text{No}\ (7 \neq \sqrt{85})\ [\text{OR}\ (b)\, \text{Proved non-collinear}]}$

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